

Predictive Modelling of Blood Demand for Optimized Hospital Inventory Management Using Machine Learning

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Objective: To develop a machine learning-based predictive system for forecasting hospital blood demand and optimizing inventory management, aiming to reduce shortages and wastage.

Problem Context and Claimed Contributions:

The authors frame the problem as follows:

- Blood is a perishable, critical resource with fluctuating demand driven by factors like emergencies, seasonal trends, and demographic shifts.
- Traditional forecasting methods (e.g., manual estimation, basic time-series) fail to capture these dynamic, non-linear patterns, leading to inefficiencies such as stockouts or expired units.
- Existing ML approaches in healthcare logistics are often siloed, either focusing only on prediction or only on optimization, without a unified decision-support framework.

To address this, they propose a hybrid machine learning and optimization framework that:

1. Combines feature-based regression (XGBoost) and time-series forecasting (Facebook Prophet) into a single, more accurate hybrid model.
2. Integrates the predictions into a simulation-based inventory optimization module that considers shelf-life, safety stock, and reorder points.
3. Delivers the output through an interactive dashboard (e.g., Power BI) for real-world decision support.
4. It is validated using real hospital data from Zimbabwe.

Used Dataset:

- Source: National Blood Services Zimbabwe and partner hospitals.
- Content: Historical transfusion records, inventory logs, and anonymized patient data.
- Key Variables: Daily blood usage, blood type, patient demographics, day-of-week, month, number of accident cases, number of maternity cases.
- Inclusion Criteria: Hospitals with 2–5 years of digitized records, varied demand patterns (urban/rural), and structured, anonymized data.
- Exclusion Criteria: Paper-based records, non-hospital blood banks, datasets with PII.

Methodology:

- 1. Data Preprocessing: Cleaning, feature engineering (temporal features, lags, rolling statistics), time-series decomposition, normalization.
- 2. Predictive Modelling:
 - Models: XGBoost (for feature-based prediction) and Facebook Prophet (for capturing seasonality).
 - Hybridization: Simple averaging of predictions from both models.
 - Validation: Time-series cross-validation; evaluation using MAE, RMSE, R².
- 3. Inventory Optimization & Simulation:
 - Forecasts fed into an optimization module calculating dynamic reorder points and safety stock.
 - A discrete-event simulation model tested inventory policies under various demand scenarios.
- 4. Evaluation Metrics:
 - Predictive Accuracy: MAE, RMSE, R².
 - Operational Performance: Stock-out rate, turnover ratio, percentage of expired units.

Results:

Model	RMSE	MAE	R ² Score	Key Strength
XGBoost	9.72	7.42	0.73	Captures non-linear feature relationships (e.g., accidents, maternity cases).
Facebook Prophet	Reported lower than hybrid	Reported lower than hybrid	Reported lower than hybrid	Captures weekly/yearly seasonality effectively.
Hybrid (XGBoost + Prophet)	12.41	9.78	0.85	Combines strengths of both; highest overall accuracy.

Conclusion:

The proposed hybrid ML and optimization framework provides a robust, data-driven solution for blood demand forecasting and inventory management. It outperforms individual models and offers a practical decision-support tool to enhance operational efficiency, reduce costs, and improve patient care in hospital blood supply chains.

Future Direction:

- Incorporate real-time external data (e.g., weather, public events, donor availability) to improve forecast accuracy.
- Explore advanced neural architectures (e.g., LSTM, Transformers) for modelling more complex temporal dependencies.
- Test the model's generalizability across different geographic regions and healthcare systems.
- Develop a lightweight, real-time deployment variant for integration directly into hospital management systems.
- Extend the framework to consider multi-echelon (hospital-regional-national) blood supply chain optimization.

Integrating machine learning and encryption for effective data management in blood bank supply chains

Authors: K. M. Kirupa Shankar, V. Santhi

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Objective: To develop a secure, integrated framework for cloud-based healthcare data management that simultaneously ensures high-accuracy classification of sensitive data, efficient encryption, and verifiable data integrity, specifically tailored for blood bank supply chains.

Problem Context and Claimed Contributions:

The authors frame the problem as follows:

- Healthcare data, especially in blood banks, is highly sensitive, time-critical, and subject to strict privacy regulations (e.g., HIPAA).
- Existing cloud-based solutions often address security, classification, or efficiency in isolation, creating fragmented systems. Common limitations include:
 - High encryption latency making real-time processing infeasible.
 - Lack of integrated auditability and third-party verification (TPA).

- Absence of intelligent feature selection to pre-filter sensitive data before cloud transmission.

To address these gaps, they propose a novel, holistic framework that integrates:

1. Host-Proof Storage Feature Selector: Dynamically identifies and filters sensitive attributes from datasets before cloud upload.
2. BTGO-Optimized XGBoost Classifier: Uses the Banyan Tree Growth Optimization (BTGO) algorithm to fine-tune XGBoost hyperparameters, significantly boosting classification accuracy for sensitive vs. non-sensitive data.
3. AES-GCM Encryption: Provides authenticated encryption for confidential data with minimal latency.
4. Third-Party Auditor (TPA) with SVM-based Verification: Ensures data integrity and authenticity during cloud storage and retrieval via a challenge-response mechanism.

The claimed novelty is the end-to-end integration of these components into a single, scalable, and regulation-compliant pipeline optimized for real-time blood bank operations.

Used Dataset:

- Source: Real-world, anonymized blood bank healthcare data (donor information, blood type, donation history, etc.).
- Key Features Engineered: Data Type, Null Percentage, Unique Percentage, Categorical nature, Correlation, Sensitivity.
- Analysis: Performance was evaluated across feature subsets (2 to 6 features) to demonstrate the impact of the proposed feature selector.

Methodology:

1. Data Preprocessing: Min-Max normalization.
2. Feature Selection: Novel "Host-Proof Storage Feature Selector" extracts and filters sensitive features.
3. Classification: XGBoost classifier, with hyperparameters optimized using the Banyan Tree Growth Optimization (BTGO) algorithm.
4. Encryption: Sensitive data is encrypted using the AES-GCM (Advanced Encryption Standard - Galois/Counter Mode) protocol.
5. Integrity Verification: A Third-Party Auditor (TPA) module uses an SVM-based feature matching and challenge-response mechanism to verify data integrity in the cloud without accessing plaintext.
6. Evaluation Metrics:
 - Classification: Accuracy, Precision, Recall, F1-Score, Correctness.
 - System Performance: Encryption time, Verification time, total computation time
 - Comparative Analysis: Against baseline models (SVM, RF, DT, etc.) and ablation studies.

Results:

1. Classification Performance (with 6 features):

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	Correctness (%)
Proposed (XGBoost + BTGO)	99.72	99.61	99.45	99.53	98
Random Forest (RF)	94.77	99.16	99.07	99.12	94
Decision Tree (DT)	99.65	99.00	99.00	99.00	96
Support Vector Machine (SVM)	93.00	90.69	84.00	87.00	85

2. System Efficiency:

- Encryption Latency (AES-GCM): 0.23 seconds (average).
- Integrity Verification Time (TPA): 0.12 seconds.
- Encryption Scalability: AES-GCM + SHA-256 significantly outperformed DES and 3-DES (e.g., 6,782s vs. 34,107s for 200k rows).

3. Ablation Study Impact:

Configuration	Accuracy (%)	Tenc T enc (s)	Tverify T verify (s)
Full Proposed System	99	0.23	0.12
Without BTGO Optimization	95	0.34	0.20
Without AES-GCM Encryption	93	N/A	0.32
Without TPA Verification	97	0.36	N/A

Conclusion:

The proposed integrated framework successfully bridges the gap between high-accuracy machine learning classification and robust, efficient security for cloud-based healthcare data. The BTGO-optimized XGBoost classifier achieves state-of-the-art accuracy (~99.7%), while the AES-GCM encryption and TPA verification provide low-latency security and auditability. This makes the system a practical, scalable, and regulation-compliant solution for managing sensitive data in critical environments like blood bank supply chains.

Future Direction:

- Extend the framework to handle multi-source and streaming data (e.g., from IoMT devices).
- Incorporate Federated Learning to enable decentralized model training without sharing raw data.
- Integrate Blockchain technology for immutable audit trails.
- Explore Differential Privacy techniques to provide stronger mathematical privacy guarantees.
- Adapt the system for edge computing deployments to reduce cloud dependency and latency further.

Enhancing Blood Supply Chain Management Through Time-Series Forecasting: A Gated Recurrent Unit Model

Authors: Agus Mansur, Achmad Bauravindah

Journal: Mathematical Modelling of Engineering Problems, Vol. 12, No. 6, 2025, pp. 2051-2060

DOI: <https://doi.org/10.18280/mmep.120620>

Objective: To develop and evaluate a Gated Recurrent Unit (GRU) deep learning model for accurate daily blood demand forecasting, aiming to optimize inventory management, reduce shortages, and minimize wastage in the blood supply chain.

Problem Context and Claimed Contributions:

The authors frame the problem as follows:

- Blood is a perishable resource with a complex, non-linear, and volatile demand pattern influenced by seasonality, emergencies, and demographic shifts.

- Traditional forecasting methods (e.g., Moving Averages, ARIMA) lack the flexibility to capture these complex temporal dependencies and sudden changes, leading to inefficiencies.
- There is a critical need for accurate, real-time forecasting tools to address blood shortages (exemplified by a 25% drop in Indonesia's supply) and reduce the waste of expired units.

To address this, they propose a GRU-based deep learning model specifically for time-series forecasting of blood donations. The claimed contributions are:

1. Application of GRU to Blood Supply: Implements a 4-layer GRU network, a variant of RNNs known for efficiency in capturing sequential patterns, to forecast daily blood donation demand.
2. End-to-End Forecasting Pipeline: Provides a complete methodology from data preprocessing (chronological alignment, Min-Max normalization, sliding window creation) to model training and evaluation.
3. Empirical Validation on Real Data: Demonstrates the model's effectiveness using a real-world dataset from the Indonesian Red Cross (PMI) in Yogyakarta.
4. Focus on Practical Inventory Impact: Positions the accurate forecasts as a direct tool for proactive inventory control, safety stock adjustment, and reorder point optimization.

Used Dataset:

- Source: Indonesian Red Cross (Palang Merah Indonesia - PMI), Yogyakarta branch.
- Period: October 1, 2023, to January 31, 2024 (4 months).
- Initial Data: 5,170 individual donation records with 13 attributes.
- Processed Data: Aggregated into a daily time-series of 120 data points (daily donation counts). Key variables focused on donation date and blood type.
- Preprocessing: Missing values (30 points) imputed using Forward Fill (FFILL); data normalized via Min-Max scaling; structured into sequences using a sliding window approach.
- Split: 80% for training, 20% for testing, with temporal splitting to preserve chronological order.

Methodology:

1. Data Preprocessing: Chronological ordering, handling missing values (FFILL), Min-Max normalization, and creation of input sequences using a sliding window.
2. Model Architecture: A 4-layer GRU model.
 - Layers 1-3: 512 units each.
 - Layer 4: 1024 units.
 - Dropout layer (rate=0.2) after the final GRU layer to prevent overfitting.
 - Final Dense layer with 5 units.
3. Training Configuration:

- Optimizer: Adam (learning rate=0.001).
 - Loss Function: Huber Loss (robust to outliers).
 - Batch Size: 8.
 - Epochs: 100.
4. Evaluation Metrics: Mean Absolute Error (MAE) and Mean Squared Error (MSE) calculated on the normalized test data.

Results:

Metric	Value (on Normalized Test Data)	Implication
Mean Absolute Error (MAE)	0.2033	Indicates low average forecast error.
Mean Squared Error (MSE)	0.0923	Suggests the model avoids large prediction errors.

Conclusion:

The proposed GRU model proves to be an effective and accurate tool for forecasting daily blood demand. Its ability to learn complex temporal patterns from historical data demonstrates significant potential for improving blood supply chain resilience. By providing reliable forecasts, the model enables more proactive inventory management, helping to balance supply with demand, reduce the risk of shortages, and minimize waste due to expiration.

Future Direction:

- Expand Data Scope: Use longer time-series data (≥ 1 year) to better capture long-term seasonal and annual cycles.
- Improve Generalizability: Validate and potentially retrain the model using data from diverse geographic regions and demographic contexts.
- Address Overfitting Risk: Explore model simplification (parameter reduction), pruning, or more advanced regularization techniques to mitigate risks from the high parameter-to-data ratio.
- Incorporate External Variables: Integrate real-time data sources like hospital demand, emergency alerts, or public health events to make forecasts more dynamic and responsive.
- Model Comparison & Hybridization: Compare performance with other advanced models (e.g., LSTM, Transformers) or develop hybrid models to leverage different architectural strengths.

Development of blood demand prediction model using artificial intelligence based on national public big data

Authors: Hi Jeong Kwon, Sholhui Park, Young Hoon Park, Seung Min Baik, Dong Jin Park

Journal: Sagepub

Year: 2024 (accepted December 2023)

DOI: 10.1177/20552076231224245

Objective: To develop an artificial intelligence-based blood demand prediction model using national open big data and hospital transfusion records to forecast monthly blood transfusion requirements for different blood components and types.

Problem context and claimed contributions:

The authors frame the problem as follows:

- Global blood shortages are worsening due to aging populations, declining donations, and disruptions like COVID-19.
- Existing blood demand prediction studies are often limited to specific diseases, patient groups, or single institutions, lacking generalizability.
- There is a need for accurate, data-driven models to optimize blood inventory management and reduce waste.

To address this, they propose a novel AI-based prediction framework that:

- Uses national open big data (e.g., HIRA, Statistics Korea) combined with single-institution transfusion records.
- Develops separate models for different blood components (RBCs, FFP, PC) and blood types (A+, B+, O+, AB+).
- Implements ensemble learning to enhance prediction accuracy and robustness.

Used Dataset:

- Source: Health Insurance Review and Assessment Service (HIRA) and Statistics Korea (national data), plus Yeouido St. Mary's Hospital records.
- Period: January 2010 – December 2021.
- Data includes: Monthly blood inventory, transfusion demand, national donations, COVID-19 cases, hospital surgeries, crossmatch tests, antibody screens, and departmental transfusion counts.
- Blood components: Red blood cells (RBCs), fresh frozen plasma (FFP), platelet concentrate (PC).
- Blood types: A+, B+, O+, AB+ (RhD-negative excluded due to low prevalence in Korea).

Methodology:

- Preprocessing: Data cleaning, handling missing values, merging datasets based on time and patient identifiers.
- Models: Three boosting algorithms—XGBoost, LightGBM, CatBoost—were used for regression.
- Ensemble: Soft-weighted voting ensemble of the three models.
- Validation: Stratified k-fold cross-validation (k=10) to prevent data loss and ensure robustness.
- Feature Importance: Analyzed using built-in FI methods to identify key predictors.
- Metrics: R², MAE, MSE, RMSE.

Results:

Blood Component	Blood Type	Best Model	R ²	MAE	RMSE
RBC	A+	XGBoost	0.5360	23.06	27.40
RBC	O+	CatBoost	0.6364	19.58	23.58
FFP	A+	XGBoost	0.4592	34.77	45.93
FFP	AB+	CatBoost	0.4917	34.23	44.53
PC	A+	LGBM	0.6981	82.63	109.89
PC	O+	LGBM	0.8399	73.13	100.81

Conclusion:

The proposed AI-based blood demand prediction model effectively forecasts transfusion needs using publicly available national data and hospital records. The ensemble approach improved accuracy, and feature analysis revealed departmental usage patterns as critical predictors. The model can aid hospitals and blood centers in proactive inventory management.

Future Directions:

- Incorporate RhD-negative blood type data for broader applicability.
- Validate the model using multi-center and international datasets.
- Develop real-time adaptive models for dynamic health crises.
- Integrate clinical and demographic variables for finer predictions.

A mathematical model for supply chain management of blood banks in India

Authors: S. Dharmaraja, Srijan Narang, Vidyottama Jain

Journal: OPSEARCH

Year: 2020

DOI: <https://doi.org/10.1007/s12597-019-00425-9>

Objective: To propose a mathematical model for optimizing the supply chain of blood banks in India by forecasting demand, determining optimal allocation of surplus units, and selecting optimal delivery routes to reduce shortages and wastage.

Problem context and claimed contributions:

The authors frame the problem as follows:

- India faces a significant blood shortage (~1 million units annually) due to mismatches between regional supply and demand.
- Blood is a perishable commodity with stochastic demand, seasonal variations, and varying shelf lives, complicating inventory management.
- Existing models often overlook integrated solutions combining forecasting, allocation, and routing.

To address this, they propose a novel three-echelon mathematical model that:

1. Forecasts demand using Seasonal ARIMA (SARIMA).
2. Optimizes allocation from surplus to shortage banks using a Blood Transportation Allocation Problem (BTAP).
3. Determines optimal delivery routes via a Vehicle Routing Problem (VRP).

Used Dataset:

- Data Source: Monthly aggregate blood requirement data from a government hospital (1990–2014).
- Data Type: Time-series data on blood unit requirements.
- Geographic Focus: India, with state-wise statistics referenced.
- Validation: Forecasts compared against actual data from 2015–2016.

Methodology:

- Forecasting: SARIMA model selected for its ability to capture seasonality (extreme weather impacts demand).
- Allocation Model: BTAP formulated as a mixed-integer programming problem minimizing total cost (transportation, emergency supply, system costs).
- Routing Model: VRP/TSP solved using optimization techniques (branch-and-bound, cutting-plane methods).

- Validation: Compared SARIMA forecasts with two exponential smoothing methods.

Results:

Forecasting Method	Time Horizon	Performance Summary
SARIMA	1–2 years	Outperformed exponential methods, closely matched actual demand
Exponential Smoothing	>2 years	Comparable or better in some cases, but less practical for short-term planning

Example Forecast Accuracy (from Table 1):

- SARIMA predictions were consistently closer to actual monthly demand (e.g., Jan 2015: Predicted 337.05 vs. Actual 330).
- Exponential methods showed larger deviations and less seasonal sensitivity.

Conclusion:

The proposed three-tier mathematical model effectively addresses regional blood shortages by integrating forecasting, allocation, and routing. SARIMA provides reliable short-term demand predictions, while BTAP and VRP optimize logistics. The model reduces wastage and improves supply chain efficiency, though it is currently limited to whole blood and not individual components.

Future Directions:

- Extend the model to blood components (e.g., platelets, plasma) with shorter shelf lives.
- Incorporate Markov dynamic programming or simulation for stochastic demand.
- Integrate AI-based forecasting methods for improved accuracy.
- Include additional cost factors: production, outdated, inventory, shortage, and mismatch costs.
- Explore real-time tracking and dynamic routing for emergency supplies.